



FORT STREET HIGH SCHOOL

Name: _____

Teacher: _____

Class: _____

2016

HIGHER SCHOOL CERTIFICATE COURSE

ASSESSMENT TASK 3: TRIAL HSC

Mathematics Extension 2

Time allowed: 3 hours
(plus 5 minutes reading time)

Syllabus Outcomes	Assessment Area Description and Marking Guidelines	Questions
	Chooses and applies appropriate mathematical techniques in order to solve problems effectively	1-10
E2, E3	Applies appropriate strategies to construct arguments and proofs in the areas of complex numbers and polynomials	11, 12
E4, E6	Uses efficient techniques for the algebraic manipulation of conic sections and determining features of a wide variety of graphs	13
E7, E8	Applies further techniques of integration, such as slicing and cylindrical shells, integration by parts and recurrence formulae, to problems	14
E5	Uses ideas and techniques of calculus to solve problems in mechanics involving resolution of forces, resisted motion and circular motion	15
E2-E8	Synthesises mathematical processes to solve harder problems and communicates solutions in an appropriate form	16

Total Marks 100

Section I 10 marks

Multiple Choice, attempt all questions.
Allow about 15 minutes for this section.

Section II 90 Marks

Attempt Questions 11-16.
Allow about 2 hours 45 minutes for this section.

General Instructions:

- Questions 11-16 are to be started in a new booklet.
- The marks allocated for each question are indicated.
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations.
- Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used.

Section I	Total 10	Marks
Q1-Q10	/10	
Section II	Total 90	Marks
Q11	/15	
Q12	/15	
Q13	/15	
Q14	/15	
Q15	/15	
Q16	/15	
	Percent	

Section I

1 What is $-\sqrt{3} + i$ expressed in modulus-argument form?

- (A) $\sqrt{2}(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$
(B) $2(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$
(C) $\sqrt{2}(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6})$
(D) $2(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6})$

2 Consider the hyperbola with the equation $\frac{x^2}{4} - \frac{y^2}{3} = 1$.

What are the coordinates of the vertex of the hyperbola?

- (A) $(\pm 2, 0)$ (B) $(0, \pm 2)$
(C) $(0, \pm 4)$ (D) $(\pm 4, 0)$

3 The points $P(a \cos \theta, b \sin \theta)$ and $Q(a \cos \phi, b \sin \phi)$ lie on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the chord PQ subtends a right angle at $(0, 0)$. Which of the following is the correct expression?

- (A) $\tan \theta \tan \phi = -\frac{b^2}{a^2}$
(B) $\tan \theta \tan \phi = -\frac{a^2}{b^2}$
(C) $\tan \theta \tan \phi = \frac{b^2}{a^2}$
(D) $\tan \theta \tan \phi = \frac{a^2}{b^2}$

4 What is the value of $\int -\sec x dx$? Use the substitution $t = \tan \frac{x}{2}$.

- (A) $\ln |(t-1)(t+1)| + c$ (B) $\ln \left| \frac{1+t}{1-t} \right| + c$
(C) $\ln |(1-t)(t+1)| + c$ (D) $\ln \left| \frac{t-1}{t+1} \right| + c$

5 What is the volume of the solid formed when the region bounded by the curves $y = 2x^3$ and $y = 2\sqrt{x}$ is rotated about the x -axis? Use the method of slicing.

- (A) $\frac{5\pi}{14}$ cubic units
- (B) $\frac{10\pi}{14}$ cubic units
- (C) $\frac{5\pi}{7}$ cubic units
- (D) $\frac{10\pi}{7}$ cubic units

6 A particle of mass m is moving in a straight line under the action of a force.

$$F = \frac{m}{x^3}(6 - 10x)$$

Which of the following is an expression for its velocity at any position, if the particle starts from rest at $x = 1$?

- (A) $v = \pm \frac{1}{x} \sqrt{(-3 + 10x - 7x^2)}$
- (B) $v = \pm x \sqrt{(-3 + 10x - 7x^2)}$
- (C) $v = \pm \frac{1}{x} \sqrt{2(-3 + 10x - 7x^2)}$
- (D) $v = \pm x \sqrt{2(-3 + 10x - 7x^2)}$

7 The equation $y^3 - xy + x^3 = 7$ implicitly defines y in terms of x .

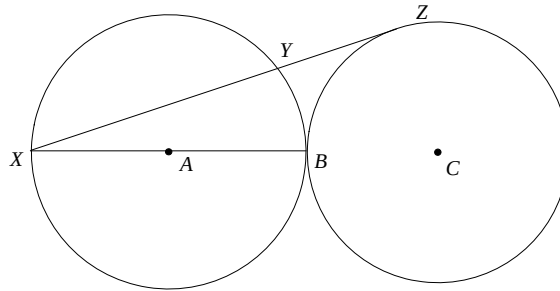
Which of the following is an expression for $\frac{dy}{dx}$?

- (A) $\frac{-3x^2}{3y^2 - 1}$
- (B) $\frac{y - 3x^2}{3y^2 - x}$
- (C) $\frac{y - 3x^2 + 7}{3y^2 - x}$
- (D) $\frac{3y^2 - y + 3x^2}{x}$

8 Let α , β and γ be roots of the equation $x^3 + 3x^2 + 4 = 0$. Which of the following polynomial equations have roots α^2 , β^2 and γ^2 ?

- (A) $x^3 - 9x^2 - 24x - 4 = 0$
- (B) $x^3 - 9x^2 - 12x - 4 = 0$
- (C) $x^3 - 9x^2 - 24x - 16 = 0$
- (D) $x^3 - 9x^2 - 12x - 16 = 0$

9 Two equal circles touch externally at B . XB is a diameter of one circle. XZ is the tangent from X to the other circle and cuts the first circle at Y .



Which is the correct expression that relates XZ to XY ?

- (A) $3XZ = 4XY$
- (B) $XZ = 2XY$
- (C) $2XZ = 3XY$
- (D) $2XZ = 5XY$

10 Which conic has eccentricity $\frac{\sqrt{13}}{3}$?

- (A) $\frac{x^2}{3} + \frac{y^2}{2} = 1$
- (B) $\frac{x^2}{3^2} + \frac{y^2}{2^2} = 1$
- (C) $\frac{x^2}{3} - \frac{y^2}{2} = 1$
- (D) $\frac{x^2}{3^2} - \frac{y^2}{2^2} = 1$

Section II

Question 11 (15 marks)

START A NEW BOOKLET

Marks

- a) (i) Find real numbers A and B such that

$$\frac{7x+1}{x^2-x-2} = \frac{A}{x+1} + \frac{B}{x-2}$$

2

- (ii) Hence, find $\int \frac{7x+1}{x^2-x-2} dx$

1

- b) (i) Prove $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

2

- (ii) Hence evaluate $\int_0^{\frac{\pi}{4}} \frac{1-\tan x}{1+\tan x} dx$

3

- c) If $z_1 = 3i$ and $z_2 = 1+i$, find values of

i) $|z_1 - z_2|$

1

ii) $z_1 + \overline{z_2}$

1

iii) $\frac{z_1}{z_2}$

1

- d) z is a complex number such that $\arg(z) = \frac{\pi}{3}$ and $|z| \leq 2$.

- i) Show the locus of the point P representing z in the Argand diagram.

2

- ii) Find the possible values of the principal argument of $z-i$ for z on the locus.

2

- a) Suppose α, β, γ and δ are the four roots of the polynomial equation

$$x^4 + px^3 + qx^2 + rx + s = 0$$

- i) find the values of $\alpha + \beta + \gamma + \delta$ and $\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta$ in terms of p, q, r and s . 2
- ii) show that $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = p^2 - 2q$ 2
- iii) apply the results in part (ii) to show that $x^4 - 3x^3 + 5x^2 + 7x - 8 = 0$ cannot have four real roots. 1
- b) Let $P(x) = x^5 - 10x^2 + 15x - 6$.
- i) show that $x = 1$ is a root of $P(x)$ of multiplicity three. 2
- ii) Hence, or otherwise, find the two complex roots of $P(x)$. 2
- c) If α, β, γ are the roots of $x^3 - 3x^2 + 2x - 1 = 0$ find the equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$. 2
- d) Evaluate $\int_0^2 t e^{-t} dt$. 4

a) Function $f(x)$ is defined by $f(x) = \log_e \left(\frac{2}{x-1} \right)$.

- i) State the domain of $f(x)$. 1
- ii) Write the equation of any asymptote to the graph of $y = f(x)$. 1
- iii) Find the x -intercept(s) of $y = f(x)$. 1
- iv) Is the function increasing or decreasing? Justify your answer. 2
- v) use your answers to the above questions to sketch graphs of the following.

$$\alpha) y = f(x) \quad \text{1}$$

$$\beta) y = |f(x)| \quad \text{1}$$

$$\gamma) y = \ln(f(x)) \quad \text{2}$$

b) $P\left(2p, \frac{2}{p}\right)$ is a variable point on the hyperbola $xy = 4$. The normal to the hyperbola at

P meets the hyperbola again at $Q\left(2q, \frac{2}{q}\right)$. M is the midpoint of PQ .

$$\text{i) Show that } q = \frac{-1}{p^3} \quad \text{2}$$

$$\text{ii) Show that } M \text{ has co-ordinates } \left[\frac{1}{p} \left(p^2 - \frac{1}{p^2} \right), p \left(\frac{1}{p^2} - p^2 \right) \right]. \quad \text{1}$$

$$\text{iii) Show that the locus of } M \text{ is given by the equation } -x^3y^3 = (y^2 - x^2)^2. \quad \text{3}$$

a) Let $I_n = \int_0^{\frac{\pi}{2}} (\sin x)^n dx$, where n is an integer, $n \geq 0$.

i) Using integration by parts, show that, for $n \geq 2$

4

$$I_n = \left(\frac{n-1}{n} \right) I_{n-2}$$

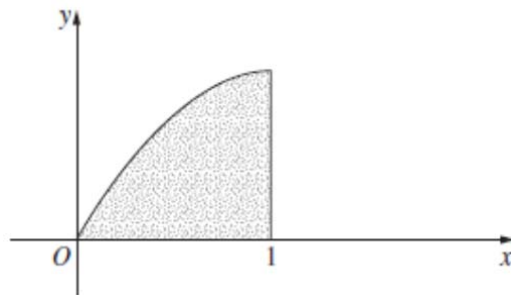
ii) Hence show that

$$\int_0^{\frac{\pi}{2}} \sin^{2n} x dx = \frac{\pi (2n)!}{2^{2n+1} (n!)^2}$$

4

b) The diagram shows the graph of $f(x) = \frac{x}{1+x^2}$ for $0 \leq x \leq 1$.

4



The area bounded by $y = f(x)$, the line $x = 1$ and the x -axis is rotated about the line $x = 1$ to form a solid.

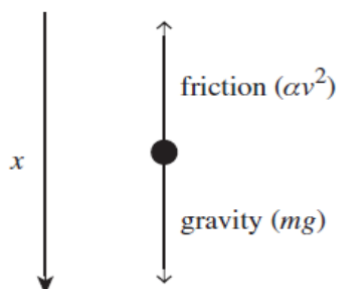
Use the method of cylindrical shells to find the volume of the solid.

c) Evaluate $\int \frac{\ln x}{(1 + \ln x)^2} dx$

3

- a) In the case of turbulent flow, the frictional force on an object travelling through a fluid is modelled as being proportional to the square of the travelling velocity.

In the case where a particle is falling vertically through the atmosphere, gravity is providing downward acceleration and the resistive frictional force is effectively acting upward, since friction always opposes motion. The situation is represented diagrammatically below.



Expressing the object's net force in terms of its velocity and a positive coefficient α yields

$$m\ddot{x} = mg - \alpha v^2$$

According to this representation, the net acceleration of the object is therefore given by

$$\ddot{x} = g - \beta v^2 \text{ where } \beta = \frac{\alpha}{m}$$

The coefficient β depends on the surface profile and mass of the object.

- i) Show that setting the initial height of the object as $x = 0$ and assuming that the initial velocity is zero yields the following equation of motion.

3

$$\ln(g - \beta v^2) = -2\beta x + \ln g$$

- ii) Show that this allows the velocity to be written as a function of position in the following way.

2

$$v(x) = \sqrt{\frac{g}{\beta}(1 - e^{-2\beta x})}$$

- iii) By using the result from (ii), state the object's terminal velocity.
- iv) Calculate the terminal velocity of a table tennis ball in free fall through the atmosphere. Take $g = 9.8 \text{ ms}^{-2}$ and $\beta = 0.2 \text{ m}^{-1}$ and give your answer in kilometres per hour.
- v) How far in metres will the table tennis ball fall before it reaches 80% of its terminal velocity?

1

1

2

b) A particle of mass m , attached by a light rod to a pivot point C , moves with constant speed in a horizontal circle whose centre O is distant h metres below C .

i. Show that the time T seconds taken for one revolution of the particle is given by

$T = 2\pi\sqrt{\frac{h}{g}}$, where g metres per second per second is the acceleration due to gravity. 3

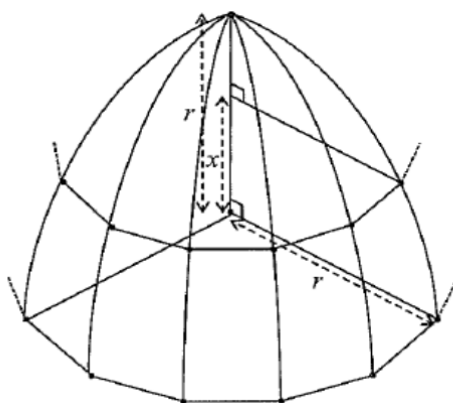
ii. discuss the effect, if any, on the motion of the particle if its speed is doubled. 3

- a) A water sprinkler is positioned at a point O on level ground at a distance of x metres from a vertical wall. Water is projected from the sprinkler at v m/s at an angle θ to the ground. Neglecting air resistance and taking g as the acceleration due to gravity. At time t , the horizontal and vertical displacement of the water from the point of projection are $x = vt \cos \theta$, and $y = vt \sin \theta - \frac{1}{2}gt^2$ respectively (DO

NOT PROVE THESE RESULTS).

- i) Show that $gx^2 \tan^2 \theta - 2xv^2 \tan \theta + 2yv^2 + gx^2 = 0$ 2
- ii) If there are two possible angles of projection, prove that $(v^2 - gy)^2 > g^2(x^2 + y^2)$. 2
- iii) show that the water from sprinkler hits the base of the wall only if $v = \sqrt{gx}$ 1
- iv) Given $v = 2\sqrt{gx}$, prove that the maximum height that can be reached by the water projected by the sprinkler up the wall is $\frac{15x}{8}$ metres. 3

- b) The diagram below shows part of a polygonal dome. Each cross-section is a regular n -sided polygon.



The vertex of the dome is r units directly above the centre of the polygonal base, which is r units from each vertex. A circular arc joins the top of the dome to each vertex of the base.

- i) Show that the area of the horizontal cross-section x units from the base is given by $\frac{n}{2}(r^2 - x^2)\sin\left(\frac{2\pi}{n}\right)$ 2
- ii) Hence show that the volume of the dome is given by $\frac{nr^3}{3}\sin\left(\frac{2\pi}{n}\right)$. 3
- iii) Show that as $n \rightarrow \infty$, the volume of the dome approaches that of a hemisphere. 2

End of Paper

Questions 1-10: Multiple Choice

1. D 2. A 3. B 4. D 5. D 6. C 7. B 8. C 9. C

Question 11:

a)

$$\text{i) } \frac{7x+1}{x^2-x-2} = \frac{A}{x+1} + \frac{B}{x-2}, \text{ hence equating numerators}$$

$$7x+1 = A(x-2) + B(x+1)$$

$$x=2: 15 = A(2-2) + B(2+1) \Rightarrow B=5$$

$$x=-1: -6 = A(-1-2) + B(-1+1) \Rightarrow A=2$$

$$\therefore \frac{7x+1}{x^2-x-2} = \frac{2}{x+1} + \frac{5}{x-2}$$

$$\text{ii) } \int \frac{7x+1}{x^2-x-2} dx$$

$$= \int \frac{2}{x+1} + \frac{5}{x-2} dx$$

$$= 2\ln(x+1) + 5\ln(x-2) + c$$

b)

$$\text{i) Let } x=a-u \quad x=0 \quad a=u$$

$$dx = -du \quad x=a \quad u=0$$

Then

10. D

Marking

Comments

❶ for correct B value

❶ for correct A value

Well done

❶ for correct integration

❶ for set-up

$$\begin{aligned} \text{ii)} \quad & \int_0^{\frac{\pi}{4}} \frac{1 - \tan x}{1 + \tan x} dx \\ &= \int_0^{\frac{\pi}{4}} \frac{1 - \tan\left(\frac{\pi}{4} - x\right)}{1 + \tan\left(\frac{\pi}{4} - x\right)} dx \\ &= \int_0^{\frac{\pi}{4}} \frac{1 - \frac{\tan \frac{\pi}{4} - \tan x}{1 + \tan \frac{\pi}{4} \tan x}}{1 + \frac{\tan \frac{\pi}{4} - \tan x}{1 + \tan \frac{\pi}{4} \tan x}} dx \quad (\text{using (i) above}) \end{aligned}$$

$$\begin{aligned}
&= \int_0^{\frac{\pi}{4}} \frac{1 + \tan \frac{\pi}{4} \tan x - \left(\tan \frac{\pi}{4} - \tan x \right)}{1 + \tan \frac{\pi}{4} \tan x} \cdot \frac{1 + \tan \frac{\pi}{4} \tan x}{1 + \tan \frac{\pi}{4} \tan x + \tan \frac{\pi}{4} - \tan x} dx \\
&= \int_0^{\frac{\pi}{4}} \frac{1 + \tan x - 1 + \tan x}{1 + \tan x + 1 - \tan x} dx \\
&= \int_0^{\frac{\pi}{4}} \frac{2 \tan x}{2} dx \\
&= \int_0^{\frac{\pi}{4}} \frac{\sin x}{\cos x} dx \\
&= \left[-\ln(\cos x) \right]_0^{\frac{\pi}{4}} \\
&= -\ln \frac{1}{\sqrt{2}} \\
&= \frac{1}{2} \ln 2
\end{aligned}$$

c)

$$\begin{aligned}
\text{i)} \quad &|z_1 - z_2| \\
&= \sqrt{(-1)^2 + 2^2} \\
&= \sqrt{5} \\
\text{ii)} \quad &z_1 + \overline{z_2} \\
&= 3i + 1 - i \\
&= 1 + 2i
\end{aligned}$$

❶ for resolving to
simple integral

❶ for value

❶ for value

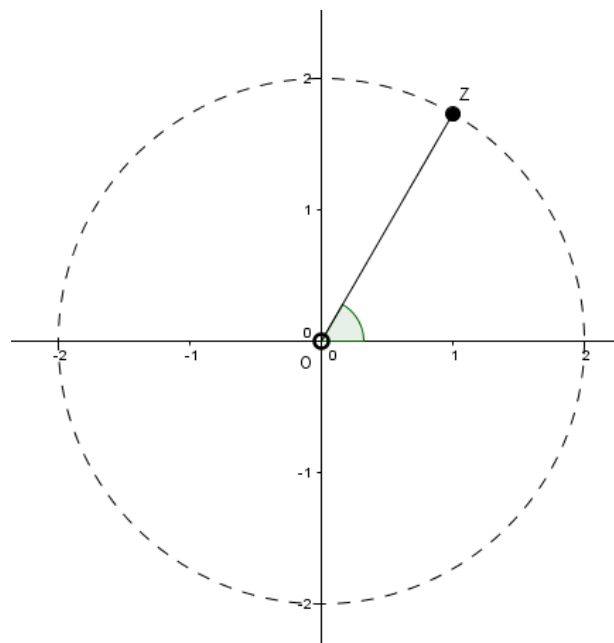
❶ for value

Well done

$$\begin{aligned}
 \text{iii) } \frac{z_1}{z_2} &= \frac{3i}{1+i} \cdot \frac{1-i}{1-i} \\
 &= \frac{3i - 3i^2}{1 - (-1)} \\
 &= \frac{3 + 3i}{2}
 \end{aligned}$$

d)

i)



The locus of P is the interval OZ , with the point at O excluded.

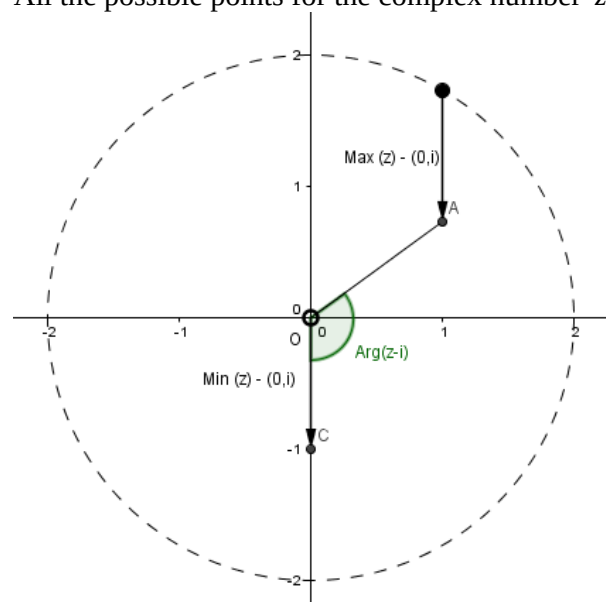
1 for value

1 for diagram

1 for explanation

Well done

ii) All the possible points for the complex number $z - i$ lie on the interval AC.



When $|z| = 2$, $z - i$ is at A. As $|z| \rightarrow 0$, $z - i \rightarrow C$. Hence:

$$\arg(C) < \arg(z - i) \leq \arg\left(A(1, \sqrt{3} - 1)\right)$$

$$\frac{-\pi}{2} < \arg(z - i) \leq \tan^{-1}(\sqrt{3} - 1)$$

❶ for value $\frac{-\pi}{2}$

❶ for value $\tan^{-1}(\sqrt{3} - 1)$

Poorly done. Most students did not get even the minimum value for $\arg(-i)$

Fort Street HS Trial HSC Extension 2 Mathematics 2016 – Solutions

Question 12:

a)

$$\begin{aligned} \text{i) } \alpha + \beta + \gamma + \delta &= \frac{-b}{a} \\ &= -p \end{aligned}$$

$$\begin{aligned} \alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta &= \frac{c}{a} \\ &= q \end{aligned}$$

$$\begin{aligned} \text{ii) } \alpha^2 + \beta^2 + \gamma^2 + \delta^2 \\ &= (\alpha + \beta + \gamma + \delta)^2 - 2(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta) \\ &= (-p)^2 - 2(q) \\ &= p^2 - 2q \end{aligned}$$

$$\begin{aligned} \text{iii) } \alpha^2 + \beta^2 + \gamma^2 + \delta^2 \\ &= p^2 - 2q \\ &= (-3)^2 - 2 \times 5 \\ &= -1 \end{aligned}$$

$$< 0$$

Hence some roots must be complex, so not all roots are real.

Marking

Comments

❶ for value

❶ for value

❶ for expansion/
connction

❶ for value

❶ for conclusion
with justification

Well done

❶ for justification

① for

$$Q(x) = x^2 + 3x + 6$$

Thus $P(x) = x^5 - 10x^2 + 15x - 6$

$$= (x-1)^3 Q(x)$$

$$= (x-1)^3 (x^2 + 3x + 6)$$

So $x^2 + 3x + 6 = 0$ gives

$$\begin{aligned}
 x &= \frac{-3 \pm \sqrt{3^2 - 4 \cdot 1 \cdot 6}}{2} \\
 &= \frac{-3 \pm \sqrt{-15}}{2} \\
 &= \frac{-3}{2} \pm i \frac{\sqrt{15}}{2} \quad \text{as the two complex roots.}
 \end{aligned}$$

c) $x = \frac{1}{\alpha}$

$$\alpha = \frac{1}{x}$$

As $\alpha^3 - 3\alpha^2 + 2\alpha - 1 = 0$

$$\left(\frac{1}{x}\right)^3 - 3\left(\frac{1}{x}\right)^2 + 2\left(\frac{1}{x}\right) - 1 = 0$$

$$1 - 3x + 2x^2 - x^3 = 0$$

$x^3 - 2x^2 + 3x - 1 = 0$ is the required equation.

❶ for values

❶ for correct method

❶ for correct equation

Well done

d) Let $u = t$ $\frac{dv}{dt} = e^{-t}$
 $\frac{du}{dt} = 1$ $v = -e^{-t}$

Then

$$\int_0^2 t e^{-t} dt$$

$$= \left[-te^{-t} \right]_0^2 - \int_0^2 -e^{-t} dt$$

$$= (-2e^{-2} - 0) - \left[-e^{-t} \right]_0^2$$

$$= -2e^{-2} - (-e^{-2} - 1)$$

$$= 1 - 3e^{-2}$$

$$\left(= \frac{e^2 - 3}{e^2} \right)$$

❶ for correct “by parts” set-up

❶ for resolving

❶ for correct answer

Well done

Question 13:

a) For $f(x) = \log_e \left(\frac{2}{x-1} \right)$:

i) Domain needs $\frac{2}{x-1} > 0$
 $x-1 > 0$
 $x > 1$

ii) $x = 1$

iii) $\ln \left(\frac{2}{x-1} \right) = 0$

$$\frac{2}{x-1} = 1$$

$$2 = x - 1$$

$$x = 3$$

Hence the x-intercept is (3,0).

iv)

$$f(x) = \ln \left(\frac{2}{x-1} \right)$$

$$= \ln 2 - \ln(x-1)$$

$$f'(x) = 0 - \frac{1}{x-1}$$

$$< 0 \quad \text{as } x > 1 \text{ from (i).}$$

Hence the function is always decreasing.

Marking

Comments

❶ for value

❶ for value

Well done for (i) –
(iii)

❶ for point

Some had errors in
 $\frac{dy}{dx}$ or didn't simplify

❶ for working

sufficiently to
support argument

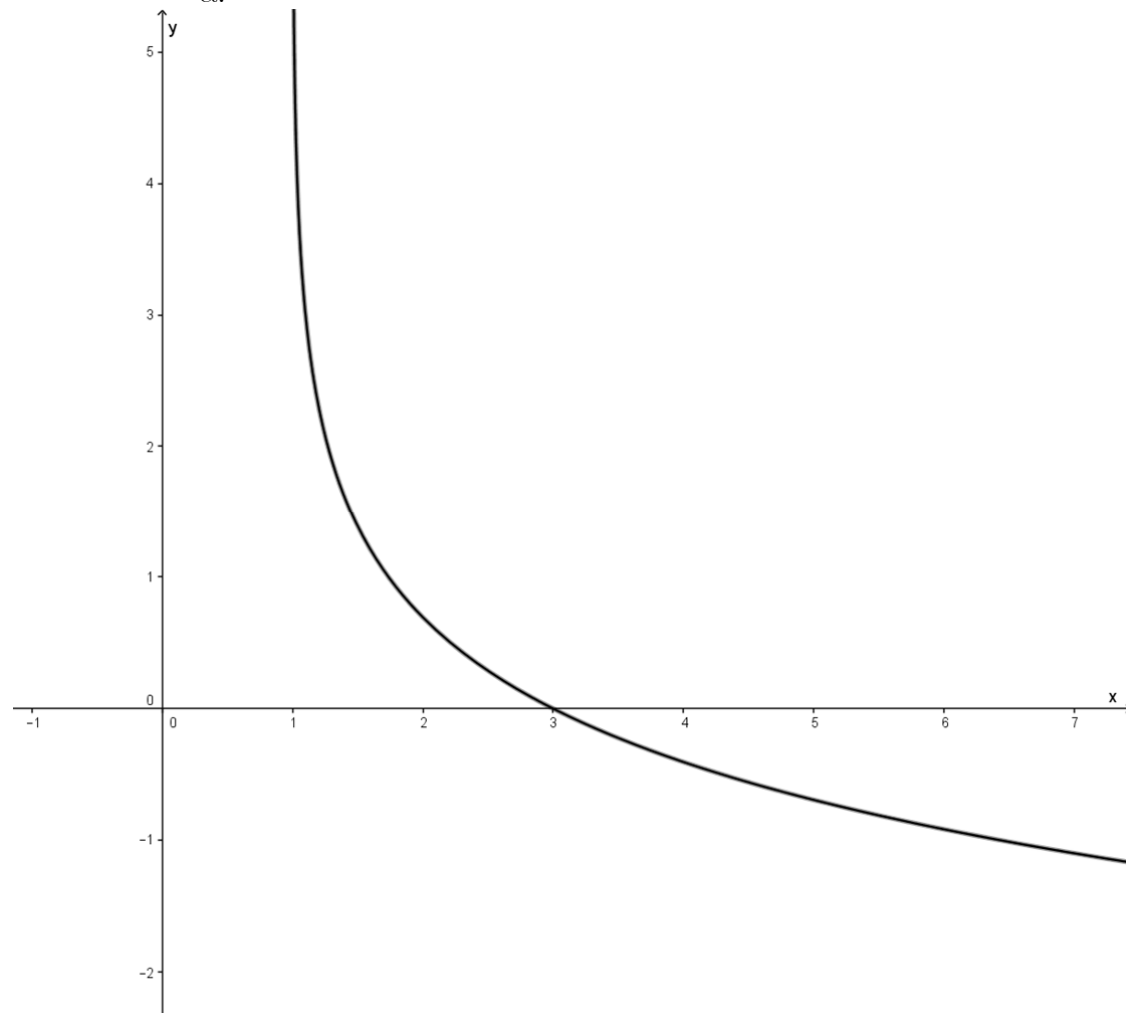
that $\frac{dy}{dx} < 0$. Some

❶ for justification

incorrect limits used
instead of sign of
 $\frac{dy}{dx}$.

v)

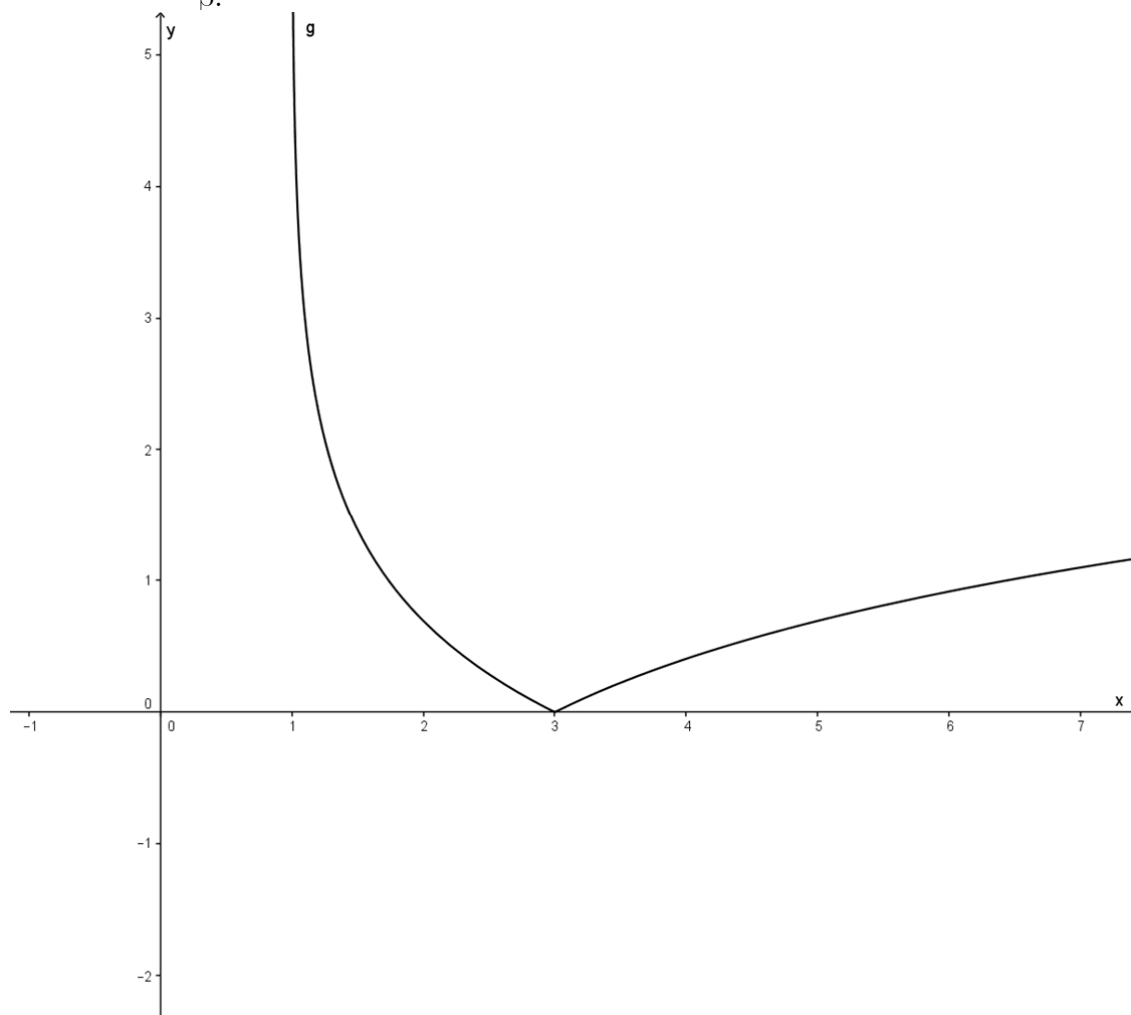
$\alpha.$



1 for graph

Well done

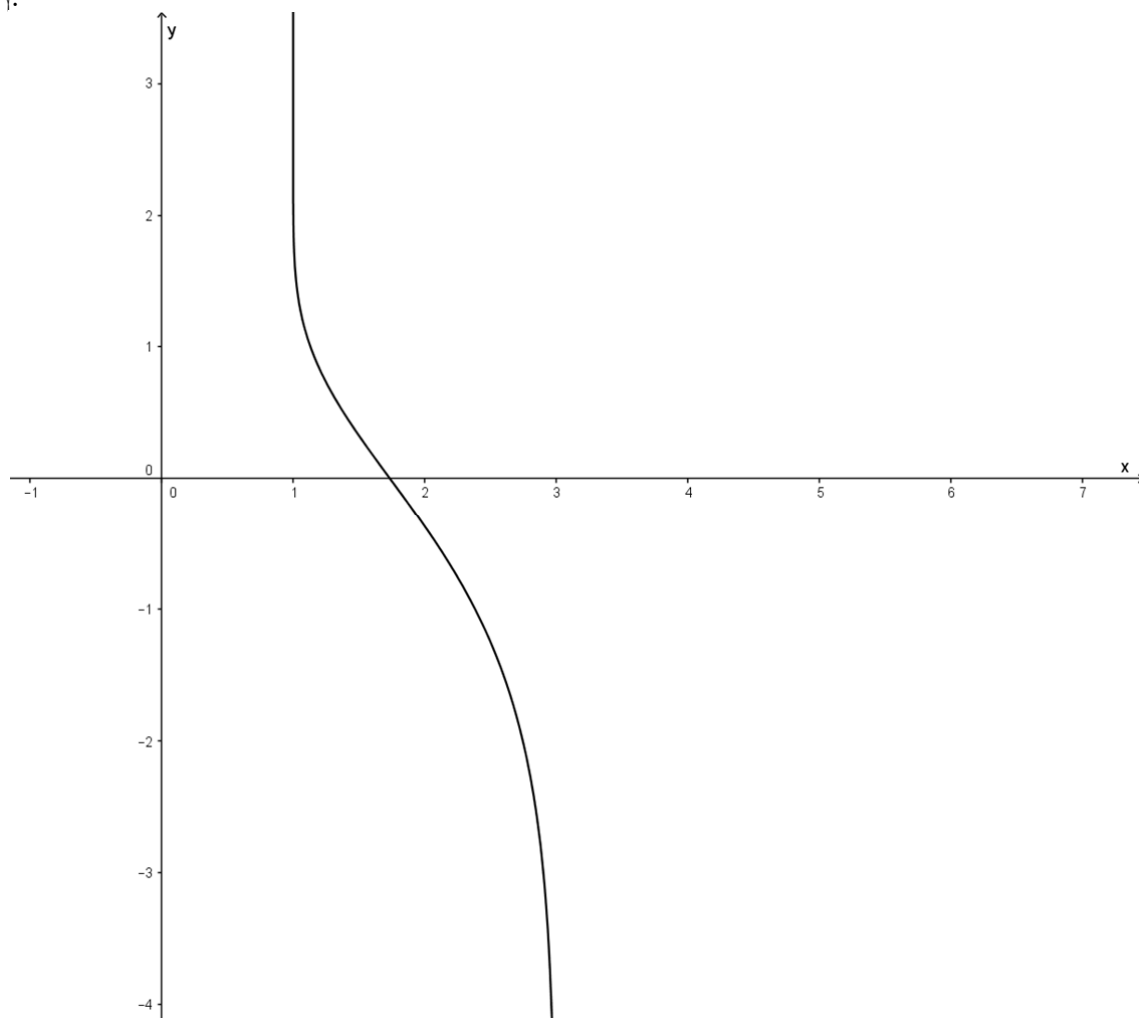
β.



1 for graph

Well done

7.



1 for graph

1 for intercept/
asymptotes

*A few did not realize
restriction on
domain of $1 < x < 3$.*

b)

$$\text{i) } y = \frac{4}{x}$$

$$\frac{dy}{dx} = \frac{-4}{x^2}$$

$$\text{At } P\left(2p, \frac{2}{p}\right):$$

$$m = \frac{-1}{p^2}, \text{ so normal has gradient } p^2. \text{ Hence}$$

Gradient of PQ:

$$m_{PQ} = \frac{\frac{2}{q} - \frac{2}{p}}{2q - 2p}$$

$$= \frac{2\left(\frac{p-q}{pq}\right)}{2(q-p)}$$

$$= \frac{-1}{pq}$$

As these are the same (equal):

$$p^2 = \frac{-1}{pq}$$

$$q = \frac{-1}{p^3}$$

(b) (i) and (ii)
Generally well done

❶ for gradient

❶ for equating

$$\begin{aligned} \text{ii) } x_m &= \frac{2q+2p}{2} & \text{and } y_m &= \frac{\frac{2}{q} + \frac{2}{p}}{2} \\ &= p+q & &= \frac{1}{p} + \frac{1}{q} \\ &= p - \frac{1}{p^3} & &= \frac{1}{p} - p^3 \\ &= \frac{1}{p} \left(p^2 - \frac{1}{p^2} \right) \end{aligned}$$

iii)

$$\begin{aligned} \frac{y}{x} &= \frac{\frac{1}{p} \left(p^2 - \frac{1}{p^2} \right)}{p \left(\frac{1}{p^2} - p^2 \right)} \\ &= p^2 \frac{\left(p^2 - \frac{1}{p^2} \right)}{- \left(p^2 - \frac{1}{p^2} \right)} \\ &= -p^2 \end{aligned}$$

$$y = -p^2 x$$

$$p^2 = \frac{-y}{x} \quad \text{①}$$

Then from

❶ for both values

❶ for y to x relationship

(iii) many students tried to use the incorrect method of showing $LHS=RHS$. This does not show the equation is the locus of M. It just shows that the identity is true (without relating it to the locus) which is not what the question required.

$y = p \left(\frac{1}{p^2} - p^2 \right)$ $y^2 = p^2 \left(\frac{1}{p^2} - p^2 \right)^2$ $y^2 = \frac{-y}{x} \left(\frac{-x}{y} - \frac{-y}{x} \right)^2 \text{ using ① above.}$ $y^2 = \frac{-y}{x} \left(\frac{y^2 - x^2}{xy} \right)^2$ $= \frac{-1}{x^3 y} (y^2 - x^2)^2$ $-x^3 y^3 = (y^2 - x^2)^2 \quad \text{as reqd.}$	① for substitution ① for algebra to result
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Question 14:

a)

$$\text{i) } I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$$

$$= \int_0^{\frac{\pi}{2}} \sin^{n-1} x \cdot \sin x \, dx$$

$$\text{Let } u = \sin^{n-1} x \quad dv = \sin x$$

$$du = (n-1)\sin^{n-2} x \cdot \cos x \quad v = -\cos x$$

Then by parts:

$$I_n = [0 - 0] + (n-1) \int_0^{\frac{\pi}{2}} (1 - \sin^2 x) \cdot \sin^{n-2} x \, dx$$

$$= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x - \sin^n x \, dx$$

$$= (n-1)(I_{n-2} - I_n)$$

Hence

$$I_n = (n-1)I_{n-2} - (n-1)I_n$$

$$(n-1)I_n + I_n = (n-1)I_{n-2}$$

$$nI_n = (n-1)I_{n-2}$$

$$I_n = \frac{(n-1)}{n} I_{n-2} \quad \text{as reqd.}$$

Marking

Comments

① for by parts set-up

Generally well done

① resolves by parts...

① resolves to I_n, I_{n-2} relationship

① resolves algebra to result

ii) To show $\int_0^{\frac{\pi}{2}} \sin^{2n} x \, dx = \frac{\pi(2n)!}{2^{2n+1}(n!)^2}$

i.e.

$$\int_0^{\frac{\pi}{2}} \sin^{2n} x \, dx = I_{2n}$$

$$= \left(\frac{2n-1}{2n} \right) I_{n-2}$$

$$= \left(\frac{2n-1}{2n} \right) \left(\frac{2n-3}{2n-2} \right) I_{n-4}$$

$$= \left(\frac{2n-1}{2n} \right) \left(\frac{2n-3}{2n-2} \right) \dots I_2$$

$$= \left(\frac{2n-1}{2n} \right) \left(\frac{2n-3}{2n-2} \right) \dots \frac{1}{2} I_0$$

Now $I_0 = \int_0^{\frac{\pi}{2}} \sin^0 x \, dx$

$$= \int_0^{\frac{\pi}{2}} 1 \, dx$$

$$= [x]_0^{\frac{\pi}{2}}$$

$$= \frac{\pi}{2}$$

From the pattern $\left(\frac{2n-1}{2n} \right) \left(\frac{2n-3}{2n-2} \right) \dots$ we note the missing terms to be in sequence in the numerator are $2n, 2n-2, 2n-4, \dots$ hence inserting these:

❶ establishes pattern

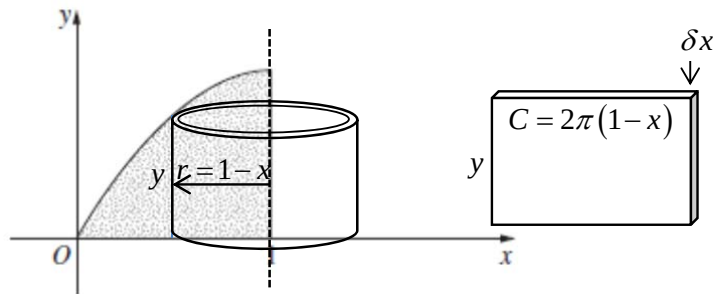
❶ finds I_0

❶ identifies missing terms to complete pattern

Many had problems establishing a pattern, and only a few found the value for I_0 .

$$\begin{aligned}
 I_{2n} &= \frac{2n}{2n} \left(\frac{2n-1}{2n} \right) \frac{2n-2}{2n-2} \left(\frac{2n-3}{2n-2} \right) \frac{2n-4}{2n-4} \dots \frac{1}{2} \cdot \frac{\pi}{2} \\
 &= \frac{2n \cdot 2n-1 \cdot 2n-2 \cdot 2n-3 \dots 2 \cdot 1 \cdot \pi}{(2n)^2 (2n-2)^2 \dots 4^2 \cdot 2^2 \cdot 2} \\
 &= \frac{\pi (2n)!}{2^2 n^2 \cdot 2^2 (n-1)^2 \cdot 2^2 (n-2)^2 \dots 2^2 2^2 \cdot 2^2 1^2 \cdot 2} \\
 &= \frac{\pi (2n)!}{(2^2)^n \cdot 2 \cdot (n(n-1)(n-2) \dots 2 \cdot 1)^2} \\
 &= \frac{\pi (2n)!}{2^{2n+1} (n!)^2} \quad \text{as reqd.}
 \end{aligned}$$

b)



Hence the shell is rectangular with area
 $\delta A = lb$

$$= 2\pi(1-x)y \quad \text{where } y = \frac{x}{1+x^2}$$

Thus the volume of the shell is

$$\delta V = 2\pi(1-x)y \cdot \delta x$$

❶ resolves factorials and powers correctly

❶ set-up: diagram with derivation of δV in appropriate detail

Set-up was generally poorly done. Radius was often incorrectly identified as x .

$$\begin{aligned}
 V &= \lim_{\delta x \rightarrow 0} \sum_0^1 \delta V \\
 &= \lim_{\delta x \rightarrow 0} \sum_0^1 2\pi(1-x)y \cdot \delta x \\
 &= 2\pi \int_0^1 (1-x)y \cdot dx \\
 &= 2\pi \int_0^1 (1-x) \frac{x}{1+x^2} \cdot dx \\
 &= 2\pi \int_0^1 \frac{x-x^2}{1+x^2} \cdot dx \\
 &= 2\pi \int_0^1 \frac{-1-x^2+1+x}{1+x^2} \cdot dx \\
 &= 2\pi \int_0^1 \frac{-(1+x^2)+1+x}{1+x^2} \cdot dx \\
 &= 2\pi \int_0^1 -1 + \frac{1}{1+x^2} + \frac{x}{1+x^2} \cdot dx \\
 &= 2\pi \left[-x + \tan^{-1} x + \frac{1}{2} \ln(1+x^2) \right]_0^1 \\
 &= 2\pi \left[\left(-1 + \tan^{-1} 1 + \frac{1}{2} \ln(1+1) \right) - \left(-0 + \tan^{-1} 0 + \frac{1}{2} \ln(1+0) \right) \right] \\
 &= \pi \left(\ln 2 + \frac{\pi}{2} - 2 \right)
 \end{aligned}$$

❶ resolves sum to integral

❶ resolves into partial fractions (any method)

❶ answer

$$c) \int \frac{\ln x}{(1 + \ln x)^2} dx$$

Let $u = 1 + \ln x$ so $\ln x = u - 1$, then

$$du = \frac{1}{x} dx \quad x = e^{u-1}$$

$$dx = e^{u-1} du$$

$$\int \frac{\ln x}{(1 + \ln x)^2} dx$$

$$= \int \frac{u-1}{u^2} \cdot e^{u-1} du$$

$$= \int \left(\frac{1}{u} - \frac{1}{u^2} \right) e^{u-1} du$$

$$= \int \frac{e^{u-1}}{u} du - \int \frac{e^{u-1}}{u^2} du$$

$$= \int \frac{e^{u-1}}{u} du - \left[-\left(\frac{e^{u-1}}{u} \right) - \int \frac{-1}{u} e^{u-1} du \right]$$

$$= \int \frac{e^{u-1}}{u} du + \left(\frac{e^{u-1}}{u} \right) - \int \frac{e^{u-1}}{u} du$$

$$= \left(\frac{e^{u-1}}{u} \right) + c$$

$$= \frac{x}{1 + \ln x} + c$$

❶ uses appropriate substitution

Generally well done

❶ resolves to simple integration

❶ answer back into original variable

Question 15:

a)

i)

$$\ddot{x} = g - \beta v^2$$

$$v \frac{dv}{dx} = g - \beta v^2$$

$$\frac{v dv}{g - \beta v^2} = dx$$

$$\frac{-2\beta v dv}{g - \beta v^2} = -2\beta dx$$

$$\ln(g - \beta v^2) = -2\beta x + c$$

When $x = 0, v = 0$:

$$-2\beta \times 0 + c = \ln(g - \beta \times 0^2)$$

$$c = \ln g$$

$$\therefore \ln(g - \beta v^2) = -2\beta x + \ln g$$

ii)

$$-2\beta x = \ln(g - \beta v^2) - \ln g$$

$$-2\beta x = \ln\left(\frac{g - \beta v^2}{g}\right)$$

$$e^{-2\beta x} = \frac{g - \beta v^2}{g}$$

$$ge^{-2\beta x} = g - \beta v^2$$

$$\beta v^2 = g - ge^{-2\beta x}$$

$$v^2 = \frac{g}{\beta}(1 - e^{-2\beta x})$$

$$v = \sqrt{\frac{g}{\beta}(1 - e^{-2\beta x})} \text{ (as } v > 0)$$

Marking

Comments

❶ resolves to integration

❶ resolves integral correctly

❶ correct c value

❶ resolves to exponentials

❶ correct algebra to result

Generally well done.
Several students successfully used

$\int_0^x \sim dx = \int_0^v \sim dv$
method.

Well done.

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iii) Terminal velocity when $x \rightarrow \infty, \Rightarrow e^{-2\beta x} \rightarrow 0$, hence

$$\begin{aligned} v_T &= \sqrt{\frac{g}{\beta}(1-0)} \\ &= \sqrt{\frac{g}{\beta}} \quad \text{as reqd.} \end{aligned}$$

$$\begin{aligned} \text{iv) } v_T &= \sqrt{\frac{g}{\beta}} \\ &= \sqrt{\frac{9.8}{0.2}} \\ &= 7 \text{ m/s} \\ &= \frac{7 \times 60 \times 60}{1000} \text{ km/h} \\ &\approx 25.2 \text{ km/h} \end{aligned}$$

v) From (i):

$$\begin{aligned} -2\beta x &= \ln g - \ln(g - \beta v^2) \\ x &= \frac{-1}{2\beta} \ln\left(\frac{g}{g - \beta v^2}\right) \\ &= \frac{-1}{2 \times 0.2} \ln\left(\frac{9.8}{9.8 - 0.2 \times (0.8 \times 7)^2}\right) \\ &\approx 2.55 \text{ m} \end{aligned}$$

① correct reasoning

① correct result

① substitutions correct

① correct answer

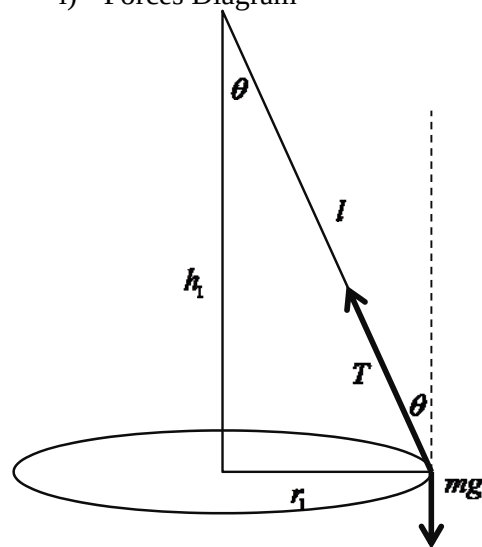
Some used the acceleration definition to get to the same result. Don't substitute ∞ into equations – it's not a number!

Some did not read requirements and left the answer in m/s.

Generally well done.

b)

i) Forces Diagram



Resolving Forces:

Vertically: no motion

$$\sum F_i = 0$$

$$T \cos \theta - mg = 0$$

$$T \cos \theta = mg \quad \textcircled{1}$$

$$\begin{aligned} \textcircled{2} \div \textcircled{1}: \tan \theta &= \frac{mr\omega^2}{mg} \\ &= \frac{r\omega^2}{g} \end{aligned}$$

But from diagram: $\tan \theta = \frac{r}{h}$, hence

Horizontally: circular motion

$$\sum F_i = mr\omega^2$$

$$T \sin \theta = mr\omega^2 \quad \textcircled{2}$$

① forces diagram and resolution of forces correct

Set-up and forces diagram, with resolution of forces, usually poorly done.

$$\frac{r}{h} = \frac{r\omega^2}{g}$$

$$\omega^2 = \frac{g}{h}$$

$$\omega = \sqrt{\frac{g}{h}}$$

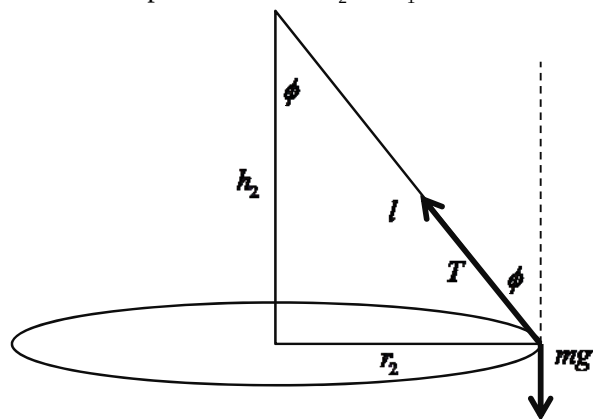
For Period:

$$T = \frac{2\pi}{\omega}$$

$$= \frac{2\pi}{\sqrt{\frac{g}{h}}}$$

$$= 2\pi\sqrt{\frac{h}{g}} \text{ as reqd.}$$

ii) When the speed doubles: $v_2 = 2v_1$



① linking the two expressions for $\tan \theta$.

① resolved for ω with correct resolution to required result.

We know from $h = \frac{g}{\omega^2}$ that h decreases as ω increases and so clearly $\phi > \theta$.

So from part (i):

$$v_1 = r_1 \omega_1$$

$$h_1 = \frac{g}{\omega_1^2}$$

$$T_1 = \frac{2\pi}{\omega_1}$$

compared to double speed:

$$v_2 = r_2 \omega_2 \text{ ①}$$

$$h_2 = \frac{g}{\omega_2^2} \text{ ②}$$

$$T_2 = \frac{2\pi}{\omega_2} \text{ ③}$$

With $v_2 = 2v_1$, this gives the following relationships:

①:

$$\begin{aligned} \omega_2 &= \frac{v_2}{r_2} \\ &= \frac{2v_1}{r_2} \\ &= \left(\frac{2r_1}{r_2} \right) \omega_1 \end{aligned}$$

②:

$$\begin{aligned} h_2 &= \frac{g}{\omega_2^2} \\ &= \frac{g}{\left(\frac{2r_1}{r_2} \right)^2 \omega_1^2} \\ &= \left(\frac{r_2}{2r_1} \right)^2 \frac{g}{\omega_1^2} \\ &= \left(\frac{r_2}{2r_1} \right)^2 h_1 \end{aligned}$$

① appropriate working to support conclusions below

Generally poorly done.
Many assumed a doubling of ω rather than v , thus not reading the question carefully, and erroneously finding that angle or radius doubled.

As $h_2 < h_1$, this leads to

$$\left(\frac{r_2}{2r_1}\right)h_1 < h_1$$

$$r_2 < 2r_1$$

③:

$$T_2 = \frac{2\pi}{\omega_2}$$

$$= \frac{2\pi}{\left(\frac{2r_1}{r_2}\right)\omega_1}$$

$$= \left(\frac{r_2}{2r_1}\right)T_1$$

Thus height and period decrease, radius and angle increase (but NOT by doubling).

Note also that

$$\tan \theta = \frac{r_1}{h_1}$$

$$\tan \phi = \frac{r_2}{h_2}$$

$$= \frac{r_2}{\left(\frac{r_2}{2r_1}\right)h_1}$$

$$= 2 \frac{r_1}{h_1}$$

$$= 2 \tan \theta$$

$$\phi = \tan^{-1}(2 \tan \theta) \text{ (i.e. the angle is not doubled)}$$

Period is specifically mentioned in the previous part, so reference should be made to this! Ensure you understand that the parts are linked!

① height, radius and angle (at least two mentioned)

① effect on period

Question 16:

a)

i) $x = vt \cos \theta$ gives $t = \frac{x}{v \cos \theta}$

Substituting this into $y = vt \sin \theta - \frac{1}{2}gt^2$ gives

$$y = v \sin \theta \cdot \frac{x}{v \cos \theta} - \frac{1}{2}g \left(\frac{x}{v \cos \theta} \right)^2$$

$$= x \tan \theta - \frac{gx^2}{2v^2 \cos^2 \theta}$$

$$2v^2 y = 2v^2 x \tan \theta - gx^2 \sec^2 \theta$$

$$= 2v^2 x \tan \theta - gx^2 (1 + \tan^2 \theta)$$

$$= 2v^2 x \tan \theta - gx^2 - gx^2 \tan^2 \theta$$

$$0 = gx^2 \tan^2 \theta - 2v^2 x \tan \theta + gx^2 + 2v^2 y \text{ as reqd.}$$

ii) Considering the equation as a quadratic in $\tan \theta$, for two possible angles $\Delta > 0$.
Hence

$$b^2 - 4ac > 0$$

$$(2v^2 x)^2 - 4 \cdot gx^2 \cdot (2v^2 y + gx^2) > 0$$

$$4v^4 x^2 - 8gx^2 v^2 y - 4g^2 x^4 > 0$$

$$v^4 - 2gv^2 y - g^2 x^2 > 0$$

$$v^4 - 2gv^2 y > g^2 x^2$$

$$v^4 - 2gv^2 y + g^2 y^2 > g^2 x^2 + g^2 y^2$$

$$(v^2 - gy)^2 > g^2 (x^2 + y^2)$$

Marking

Comments

① resolves for t and substitutes

① links $\sec \theta$ to $\tan \theta$

① resolves to required eqn.

① link to discriminant and set-up of inequality

① correct algebra to result

Generally well done.

Some students did not realize to use the discriminant.

iii) To hit the base of the wall substitute $y = 0$ in (ii):

$$(v^2 - 0)^2 > g^2(x^2 + 0^2)$$

$$v^4 > g^2(x^2)$$

$$v > \sqrt{gx}, \text{ noting } g, x, v \text{ are all } > 0.$$

Hence $v = \sqrt{gx}$ to hit the base of the wall.

iv) Substituting $v = 2\sqrt{gx}$ into (ii):

$$\left((2\sqrt{gx})^2 - gy\right)^2 > g^2(x^2 + y^2)$$

$$(4gx - gy)^2 > g^2(x^2 + y^2)$$

$$g^2(4x - y)^2 > g^2(x^2 + y^2)$$

$$16x^2 - 8xy + y^2 > x^2 + y^2$$

$$16x^2 - 8xy > x^2$$

$$15x^2 > 8xy$$

$$y < \frac{15x}{8} \quad (\text{as } x > 0)$$

ALTERNATE METHOD (used by many)

Substituting $v = 2\sqrt{gx}$ into (i):

$$0 = gx^2 \tan^2 \theta - 8gx^2 \tan \theta + 8gxy + gx^2$$

$$= x \tan^2 \theta - 8x \tan \theta + 8y + x$$

$$y = x \tan \theta - \frac{x}{8} \tan^2 \theta - \frac{x}{8} \quad \text{①}$$

$$\therefore \frac{dy}{d\theta} = x \sec^2 \theta - \frac{x}{4} \tan \theta \sec^2 \theta$$

$$\frac{dy}{d\theta} = 0 \text{ gives}$$

❶ correct reasoning

Many students did not use the inequality derived in (ii).

❶ correct inequality

❶ correct algebra to eliminate y^2

Having missed the link earlier, many did not use this method, but used the alternate method below.

❶ correct reasoning

ALTERNATE MARKING

❶ finding $\tan \theta = 4$

$$0 = x \sec^2 \theta - \frac{x}{4} \tan \theta \sec^2 \theta$$

$$= \frac{x}{4} \sec^2 \theta (4 - \tan \theta)$$

Hence $\tan \theta = 4$ as $\sec^2 \theta \neq 0$.

$\tan \theta = 4$ into ①:

$$y = 4x - \frac{x}{8} \cdot 16 - \frac{x}{8}$$

$$= \frac{15x}{8} \quad \text{as reqd.}$$

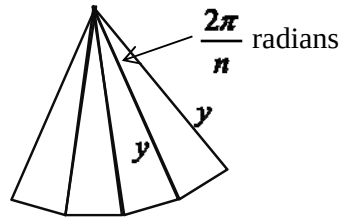
Also:

θ	$\tan^{-1}(3)$	$\tan^{-1}(4)$	$\tan^{-1}(5)$
y'	+	0	-

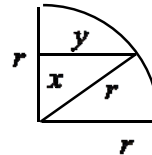
Hence the value is a maximum.

b)

i) Top view:



Side view



From the side view: $y = \sqrt{r^2 - x^2}$

① proving the result.

Many students did not confirm that the required value was a maximum.

① showing a max value.

Many students did not demonstrate the relationship between x, y and r in a right triangle. Most students recognised the need to use

$$A = \frac{1}{2} ab \sin C.$$

From the top view, one triangle has

$$A = \frac{1}{2}ab \sin C$$

$$= \frac{1}{2}y^2 \sin\left(\frac{2\pi}{n}\right)$$

So the total cross-sectional area is

$$A_r = \frac{n}{2}y^2 \sin\left(\frac{2\pi}{n}\right)$$

$$= \frac{n}{2}(r^2 - x^2) \sin\left(\frac{2\pi}{n}\right)$$

ii) $\delta V = A_r \delta x$

$$= \frac{n}{2}(r^2 - x^2) \sin\left(\frac{2\pi}{n}\right) \delta x$$

Hence

$$V = \sum \delta V$$

$$= \lim_{\delta x \rightarrow 0} \sum_0^r \frac{n}{2}(r^2 - x^2) \sin\left(\frac{2\pi}{n}\right) \delta x$$

$$= \int_0^r \frac{n}{2}(r^2 - x^2) \sin\left(\frac{2\pi}{n}\right) dx$$

$$= \frac{n}{2} \sin\left(\frac{2\pi}{n}\right) \left[r^2 x - \frac{1}{3} x^3 \right]_0^r$$

$$= \frac{n}{2} \sin\left(\frac{2\pi}{n}\right) \left[\left(r^3 - \frac{1}{3} r^3 \right) - (0) \right]$$

$$= \frac{n}{2} \cdot \frac{2r^3}{3} \cdot \sin\left(\frac{2\pi}{n}\right)$$

$$= \frac{nr^3}{3} \sin\left(\frac{2\pi}{n}\right) \quad \text{as reqd.}$$

❶ diagrams and set-up for one cross-section

❶ resolves to given result

❶ derivation of δV leading to integral

❶ resolves for r values

❶ resolves to result

Well done.

iii) Noting that $n \rightarrow \infty$ implies $\frac{2\pi}{n} \rightarrow 0$, then

$$\begin{aligned} & \lim_{n \rightarrow \infty} \frac{nr^3}{3} \sin\left(\frac{2\pi}{n}\right) \\ &= \lim_{\substack{\frac{2\pi}{n} \rightarrow 0 \\ n}} \frac{nr^3}{3} \sin\left(\frac{2\pi}{n}\right) \\ &= \lim_{\substack{\frac{2\pi}{n} \rightarrow 0 \\ n}} \frac{nr^3}{3} \cdot \frac{2\pi}{n} \cdot \frac{\sin\left(\frac{2\pi}{n}\right)}{\frac{2\pi}{n}} \\ &= \frac{2\pi r^3}{3} \lim_{\substack{\frac{2\pi}{n} \rightarrow 0 \\ n}} \frac{\sin\left(\frac{2\pi}{n}\right)}{\frac{2\pi}{n}} \\ &= \frac{2\pi r^3}{3} \cdot 1 \\ &= \frac{2\pi r^3}{3} \\ &= \frac{1}{2} \left(\frac{4}{3} \pi r^3 \right) \end{aligned}$$

which is half the volume of a sphere (a hemisphere) as reqd.

❶ resolves limit correctly

❶ demonstrates conclusion with supporting working

Hardly any students realized that as $n \rightarrow \infty$, $\frac{2\pi}{n} \rightarrow 0$ and hence were unable to correctly resolve $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.